

Logistics Object Detection & Reasoning API

PRE-HACKATHON SCREENING PRESENTATION • TRACK: COMPUTER VISION + APPLIED ML

Presenter: Aadithya R | **Backbone:** RT-DETR-Large | **Stack:** FastAPI + Docker + Terraform + AWS EC2

Live Production URL: <http://13.233.255.22>

Key Innovations: 5-class ontology (3 non-COCO), 4-stage post-processing guardrails, deterministic intent routing, and 3-tier confidence floor.

1. Domain & Supported Class Ontology

- **Industrial Domain:** Warehouse Material Handling & Freight Hub Perception.
- **5-Class Ontology:** cardboard box (0), forklift (1), freight container (2), wood pallet (3), truck (4).
- **Non-COCO Compliance:** *Cardboard box*, *freight container*, and *wood pallet* are custom non-COCO classes.
- **Operational Purpose:** Real-time inventory auditing, cargo volume estimation, and workplace safety monitoring.

2. Dataset & Split Strategy

- **Source:** Roboflow Universe Logistics Benchmark (large-benchmark-datasets/logistics-sz9jr).
- **Curated Benchmark Dataset (2,500 Images):**
 - **Train (60%):** 1,500 images (300 per class)
 - **Validation (20%):** 500 images (100 per class)
 - **Test (20% Held-Out):** 500 images (100 per class)
- **Programmatic Verification:** 12-point audit script (zero SHA256 collisions).

3. Held-Out Test Set Evaluation Benchmark

- **Overall Test Set Metrics:** mAP@0.5 = **0.5040 (50.40%)** | mAP@0.5:0.95 = **0.3408 (34.08%)** | Precision = **0.5734** | Recall = **0.4873**
- **Per-Class mAP@0.5 Results:**
 - *wood pallet*: 0.6708 (67.08%) | *truck*: 0.6515 (65.15%) | *forklift*: 0.6151 (61.51%)
 - *cardboard box*: 0.4144 (41.44%) | *freight container*: 0.1683 (16.83%)
- **Deployed Checkpoint:** weights/best.pt (Epoch 1 / epoch0.pt, best fitness 0.40391).

4. Post-Processing & Failure Root-Cause Analysis

4-Stage Post-Processing Pipeline: Class-aware NMS (0.45) $\rightarrow$ Same-class IoS containment (0.65) $\rightarrow$ Cross-class high-IoU suppression (0.80) $\rightarrow$ Boundary artifact filter.

5 Genuine Failure Cases Diagnosed:

1. *Sub-region container duplicates* (Mitigated via IoS containment suppression).
2. *Cross-class prediction overlap* (Mitigated via cross-class IoU suppression).
3. *Long container partitioning* (Active limitation; non-overlapping boxes).
4. *Wood crate vs pallet texture ambiguity* (Active limitation; slatted wood features).
5. *Low-confidence boundary artifacts* (Mitigated via boundary filter).

5. Reasoning Layer, Guardrails & Production AWS Hosting

- **Deterministic Reasoning:** Hand-written pure Python router in `app/reasoning.py` (Zero LangChain / LLM overhead).
- **6 Supported Intents:** COUNT, PRESENCE, LIST, SPATIAL, MOST_COMMON, UNKNOWN.
- **Confidence Floor Guardrail (\$0.25\$):** Returns explicit *'Insufficient information to answer confidently'* payload when evidence is weak.
- **Production AWS Hosting:** Deployed on AWS EC2 t3.small via Terraform, Nginx, and Docker Compose at <http://13.233.255.22>.